

JASVANTH. S

AI & Machine Learning Engineer

+91 8098421779 • jasvanth1010@gmail.com • [Portfolio](#) • [GitHub](#) • [LinkedIn](#)

PROFESSIONAL SUMMARY

Results-oriented AI & ML Engineer with hands-on experience designing, training, and deploying machine learning models across classification, prediction, and generative domains. Proficient in the full ML lifecycle — from data preprocessing and feature engineering to model evaluation, optimization, and production deployment. Combines a strong mathematical foundation with practical implementation skills using PyTorch, Scikit-learn, and modern LLM frameworks.

TECHNICAL SKILLS

ML Frameworks: PyTorch, Scikit-learn, Hugging Face Transformers

AI / LLM Stack: LangChain, LangGraph, CrewAI, RAG Systems, Prompt Engineering, Groq, Ollama, GPT

Generative AI: Fine-tuning, Embeddings, Vector Databases (Pinecone, ChromaDB), Retrieval Augmented Generation

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn Pipelines, Feature Engineering

Deployment: FastAPI, Flask, Streamlit, Docker, Joblib, Render

Languages: Python (Expert), SQL, JavaScript

Tools: Jupyter Notebook, Git, Kaggle, VS Code

ML PROJECTS

Heart Disease Prediction Model [Python, Scikit-learn, Pandas, Streamlit, Joblib]

- Designed a classification pipeline using multiple supervised learning algorithms (Logistic Regression, Random Forest, XGBoost) with cross-validation and hyperparameter tuning
- Achieved high predictive accuracy on clinical features; deployed as a real-time Streamlit application
- Applied SHAP-based feature importance analysis for model interpretability

Trading Prediction System [Python, Scikit-learn, yfinance, Streamlit, Matplotlib]

- Built ML models to forecast short-term stock and cryptocurrency movements from time-series data sourced via Yahoo Finance API
- Implemented feature engineering on OHLCV data; visualized historical vs. predicted price curves for actionable investor insights

EasyML — Automated ML Platform [Python, Flask, Scikit-learn, Pandas, NumPy, Render]

- Engineered an auto-ML platform enabling non-technical users to train supervised models on uploaded CSVs with zero code
- Supported model comparison, evaluation metrics, and one-click prediction endpoints via REST API

Smart Academic Assistance (Agentic RAG) [LangChain, Llama 3, Groq, CrewAI, FastAPI]

- Implemented a dual-retrieval RAG architecture combining dense vector search with LLM generation for domain-specific QA
- Reduced retrieval latency through intelligent fallback logic and Groq-accelerated inference

RESEARCH

SmartFire AI Optimizer

- Co-authored research applying GIS-based risk modelling, real-time traffic prediction, and Mixed-Integer Programming to optimize fire station placement — presented at TNFS National Conference, VIT Chennai (Top 23 Finalists)

Health Guardian AI

- Researched an agentic wearable health system integrating multi-agent coordination, digital twins, and federated learning for privacy-preserving continuous health monitoring — submitted to ICSIEKAHE Conference

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern | [Grow Your Skills](#) • Jul 2025 – Oct 2025 • Remote

- Developed LLM-based applications and contributed to model training, evaluation, and API deployment pipelines
- Designed and optimized data preprocessing and feature engineering workflows for ML automation tasks

Data Analyst Intern | [Edunet Foundation](#) • Sep 2024 – Oct 2024 • Remote

- Analyzed 10,000+ record datasets to extract predictive signals and business insights using Python and SQL
- Optimized data collection workflows, improving data accuracy by 15%

EDUCATION

B.Tech — Artificial Intelligence & Data Science

2023 – 2027

Arjun College of Technology, Coimbatore **CGPA: 8.21**

CERTIFICATIONS

Gen AI Engineering Mastermind — Outskill (2026)

AI Agents Intensive Course — Google / Kaggle (2025)

Master Class on Data Science using Python — Udemy (2025)

Master in Data Science to become a Data Scientist — Udemy (2025)

Oracle Certified Foundations Associate — Oracle University (2025)

Green Skills and AI Foundation — Edunet Foundation – Skill4Future (2024)

ACHIEVEMENTS

- Top 23 Finalist — TNFS National Technical Paper Presentation, VIT Chennai (2026)
- Presented predictive maintenance AI system at TCE Hackathon
- Speaker: 'Data Science in Industry 4.0' — ZARAK 2K25, HCAS Tech-Fest